AutoReview - AI-Based Automotive Review & Customer Sentiment Analytics

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Submitted By

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Signature of students

Somnath Mondal

ABSTRACT

The rapid proliferation of digital automotive review platforms has made customer opinion data one of the most valuable yet underutilised assets in the automotive industry. Customers routinely express nuanced opinions about multiple aspects of a vehicle - including its engine performance, battery life, fuel mileage, safety systems, ride comfort, after-sales service, infotainment, and price - all within a single review. Conventional document-level sentiment analysis tools collapse this rich, multi-faceted feedback into a single polarity label, discarding the very granularity that makes such reviews actionable for product managers, brand teams, and dealership operators.

This project, **AutoReview**, addresses this gap by developing a complete, end-to-end **Aspect-Based Sentiment Analysis (ABSA)** system purpose-built for the automotive domain. AutoReview fine-tunes two state-of-the-art transformer language models - **BERT (bert-base-uncased)** and **XLM-RoBERTa (xlm-roberta-base)** - on domain-annotated (Review, Aspect, Sentiment) triplets to perform three-class sentiment classification (Positive, Negative, Neutral) across nine predefined automotive aspects. The system incorporates a keyword-based aspect detection pipeline, a sentence-pair encoding strategy for aspect-conditioned classification, and a fully interactive Streamlit web dashboard for single-review analysis, batch processing, model training, and evaluation.

The BERT model achieves a **test accuracy of 77.07%** and a **macro F1-score of 65.74%** on a held-out test set of 1,099 samples drawn from 10,319 labelled review-aspect pairs. The system is delivered as a production-ready, modular application with export capabilities (CSV, JSON, Excel), downloadable evaluation charts (confusion matrix, training loss curves, accuracy/F1 curves), and a configurable aspect taxonomy that requires no code changes to extend.

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Signature of Guide

INTRODUCTION

The global automotive market is one of the most competitive consumer-facing industries in the world. As online review platforms -such as CarDekho, Cardwale, Google Reviews, and manufacturer portals — accumulate millions of customer reviews annually, the ability to systematically extract and interpret customer opinions has become a strategic imperative. Unlike product categories with simple binary satisfaction metrics, vehicle ownership is a complex, multi-dimensional experience. A customer purchasing a sedan may be deeply satisfied with its engine power and build quality yet simultaneously dissatisfied with its infotainment system and after-sales service quality. These co-existing, aspect-level opinions are invisible to any system that assigns a single sentiment label per review.

Natural Language Processing (NLP) has advanced dramatically with the emergence of large pre-trained transformer models. BERT (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al. in 2019, revolutionised NLP benchmarks by enabling deep contextual understanding of text through bidirectional self-attention. XLM-RoBERTa extended this capability to 100 languages through large-scale multilingual pre-training, making it suitable for international automotive review analysis. Both models can be fine-tuned on domain-specific datasets with relatively modest labelled data requirements - a critical advantage in specialised domains like automotive ABSA where large annotated corpora are not publicly available.

AutoReview leverages these advances to build a fully automated ABSA pipeline. The system accepts raw customer review text, identifies which of the nine automotive aspects (Vehicle, Engine, Battery, Mileage, Safety, Comfort, Service, Infotainment, Price) are explicitly mentioned, and for each detected aspect, classifies the expressed sentiment using a fine-tuned transformer model. The result is a structured, per-aspect sentiment profile for every review - directly actionable for automotive OEMs, dealerships, product analysts, and customer experience teams.

The project also delivers a Streamlit-based interactive dashboard that democratises access to the underlying ABSA capability, requiring no programming knowledge from end users. The dashboard supports single-review analysis with confidence scores and keyword explanations, large-scale batch processing from uploaded files, model training on custom datasets, and a fully automatic evaluation reporting suite that loads results directly from saved checkpoint files without requiring manual data entry.

Project Overview

AutoReview is structured as a layered, modular Python application organised around a clear separation of concerns between the data, model, inference, analytics, and presentation layers. At its core, the system implements the ABSA task using a sentence-pair encoding strategy where each (review, aspect) pair is jointly encoded by the transformer model - a technique shown to produce significantly better aspect-conditioned attention than single-sequence approaches.

The application entry point is a Streamlit web dashboard (`app/main.py`) that routes user interactions to five functional pages: Single Review Analysis, Batch Analytics, Model Training, Model Evaluation, and About. All five pages share a common dark-themed UI design system with Inter typography, gradient branding, and Plotly-based interactive visualisations.

The source code (`src/`) is divided into five packages. The `data` package handles file validation, text preprocessing, and synthetic dataset generation. The `models` package contains the PyTorch dataset implementation for sentence-pair encoding, the model factory for loading HuggingFace tokenisers and models with automatic device selection (CUDA/CPU), and the HuggingFace Trainer-based fine-tuning pipeline. The `inference` package provides the keyword-based aspect extractor, the transformer sentiment classifier, and the high-level inference engine that orchestrates the full ABSA pipeline with model caching and hot-swapping. The `analytics` package computes aggregate metrics (sentiment distributions, aspect frequencies, per-aspect sentiment breakdowns) and generates Plotly charts. The `export` package handles CSV, JSON, and Excel export with the openpyxl/xlsxwriter backends.

Model checkpoints are stored under `models/checkpoints/<model_key>/` with a `best_model/` subdirectory for the production-ready fine-tuned weights. A YAML configuration file (`config.yaml`) centralises all system parameters — aspect taxonomy, keyword maps, model names, training hyperparameters, and export settings — so that the system can be adapted to new domains or aspect taxonomies without modifying application code.

The complete technology stack consists of PyTorch (>= 2.0), HuggingFace Transformers (>= 4.35), Streamlit (>= 1.28), Plotly (>= 5.17), scikit-learn (>= 1.3), pandas (>= 2.0), numpy (>= 1.24), PyYAML (>= 6.0), openpyxl, xlsxwriter, and kaleido for static image export. The application runs on Python 3.10+ and supports both Windows and Linux environments, with CUDA GPU acceleration for the training pipeline.

Dataset

The dataset used in AutoReview consists of annotated (Review, Aspect, Sentiment) triplets collected from publicly available automotive customer review sources. Each row in the dataset represents a single aspect-level opinion annotation, meaning a single review may appear in multiple rows if it expresses opinions about multiple aspects. This annotation format is the standard for ABSA tasks and directly mirrors the SemEval ABSA shared task data format.

The dataset contains **10,319 labelled rows** spanning nine automotive aspects: Vehicle, Engine, Battery, Mileage, Safety, Comfort, Service, Infotainment, and Price. Each row contains a raw review text string (variable length, averaging 40–120 words), a single aspect label from the nine-class taxonomy, and a sentiment label from the three-class label set (Positive, Negative, Neutral). The dataset was split using stratified sampling to maintain class proportions across all three subsets: 70% training (6,222 rows), 15% validation (2,998 rows), and 15% test (1,099 rows). The stratified split is applied jointly on the Sentiment column to prevent class imbalance from being amplified in any particular split.

The class distribution is inherently imbalanced, reflecting the natural bias of consumer reviews: approximately 67% of annotations carry a Positive label, 19% carry a Negative label, and 14% carry a Neutral label. This imbalance is a known characteristic of review datasets — consumers tend to purchase a vehicle before reviewing it, introducing a positive bias, and Neutral expressions are inherently ambiguous and harder to annotate consistently. The imbalance most significantly affects Neutral-class F1 scores and is identified as a primary area for future improvement through data augmentation.

The dataset also supports a built-in synthetic data generator (`SyntheticDatasetGenerator`) that creates balanced, template-based review-aspect-sentiment triplets for demonstration and testing purposes. This synthetic generator allows the system to function and be demonstrated without requiring access to proprietary annotated review data.

Objective

The primary objective of AutoReview is to build a domain-specific, production-ready Aspect-Based Sentiment Analysis system that extracts per-aspect sentiment signals from raw automotive customer review text, enabling structured, granular insight generation that is not possible with document-level sentiment tools.

The system aims to fine-tune BERT and XLM-RoBERTa on automotive-domain labelled data and achieve robust three-class sentiment classification (Positive, Negative, Neutral) across nine configurable vehicle aspects, with a target test accuracy above 75% and macro F1 above 60% on the held-out test set. The objective encompasses both modelling excellence - achieving high classification performance on minority classes such as Neutral and Negative - and engineering quality, delivering a maintainable, configurable, and extensible codebase.

A secondary objective is accessibility: the system must be operable by non-technical users through an interactive web dashboard without requiring any programming knowledge. Users must be able to analyse individual reviews, process batch datasets, and inspect model evaluation results - including training curves, confusion matrices, and per-class performance reports - entirely through the graphical interface. All evaluation results must load automatically from saved checkpoint files without requiring users to re-upload datasets or re-run evaluation pipelines.

A tertiary objective is reproducibility: all model configurations, training hyperparameters, and evaluation results must be persisted to disk in human-readable formats (JSON, YAML) so that training runs can be audited, compared, and reproduced. The checkpoint management system must automatically resolve and load the best available fine-tuned weights for inference, falling back gracefully to pretrained HuggingFace Hub weights when no fine-tuned checkpoint is available.

Scope

AutoReview is scoped to explicit, keyword-detectable aspect mentions in English-language automotive customer reviews. The system detects aspects when one or more predefined keywords associated with that aspect appear as whole words or phrases in the review text. This covers the vast majority of direct, explicit opinions expressed in automotive reviews and provides a deterministic, interpretable, and configurable extraction mechanism that requires no additional labelled data for the aspect detection component.

The system classifies sentiment at the aspect level using a three-class label set - Positive, Negative, and Neutral — which aligns with the standard ABSA benchmark formulation and captures the full range of consumer opinion polarity relevant to automotive business decisions. The nine aspects in the taxonomy (Vehicle, Engine, Battery, Mileage, Safety, Comfort, Service, Infotainment, Price) were selected to cover the most commonly discussed dimensions of vehicle ownership based on review corpus analysis and are fully configurable through `config.yaml`.

The scope intentionally excludes implicit aspect detection, where an aspect is implied by the review content but not explicitly named (e.g., "It broke down three times" implicitly references Service without mentioning the word). Implicit aspect detection requires neural sequence labelling models and falls outside the current system scope. Similarly, fine-grained emotion recognition (beyond Positive/Negative/Neutral), sarcasm and irony detection, cross-sentence opinion aggregation, and multilingual input processing (for BERT; XLM-R natively supports 100 languages but training data is English-only) are outside the current scope.

The system is scoped for analytical and decision-support use - it functions as an intelligence tool to surface patterns and trends from large review corpora, not as a replacement for human judgment in product or service decisions. Confidence scores are model probability estimates and not guaranteed correctness measures. The system scope includes both individual review inference and large-scale batch processing, with structured export of results for downstream analytics tools.

Methodology

The AutoReview system follows a sequential pipeline that converts raw automotive reviews into aspect-level sentiment predictions through five stages: preprocessing, aspect extraction, classification, fine-tuning, and evaluation.

Text Preprocessing: Reviews are cleaned using the ReviewPreprocessor through HTML decoding, Unicode NFKC normalization, removal of control characters, whitespace normalization, and trimming. Conservative preprocessing is used to preserve the contextual information required by Transformer models.

Aspect Extraction: The AspectExtractor uses keyword-based regular expressions with curated automotive terms stored in config.yaml. It identifies the presence of the nine defined aspects using word-boundary matching and returns the detected aspects deterministically.

Sentiment Classification: For each detected aspect, the system creates a sentence pair in the form [CLS] aspect [SEP] review [SEP]. The Transformer processes this aspect-conditioned input, and a Linear(768, 3) classification layer predicts Positive, Negative, or Neutral sentiment. Softmax probabilities provide the prediction confidence.

Model Fine-Tuning: BERT and XLM-RoBERTa are fine-tuned using the HuggingFace Trainer API. The dataset is split into 70% training, 15% validation, and 15% testing using stratified sampling. Training uses AdamW, a learning rate of 2e-5, batch size 16, weight decay 0.01, and 5 epochs. The best model is selected based on validation macro F1-score.

Evaluation: Performance is measured using accuracy, precision, recall, macro F1-score, classification reports, and confusion matrices. The Streamlit dashboard displays training loss, validation performance, and confusion matrices using Plotly, with charts available for high-resolution download.

Result

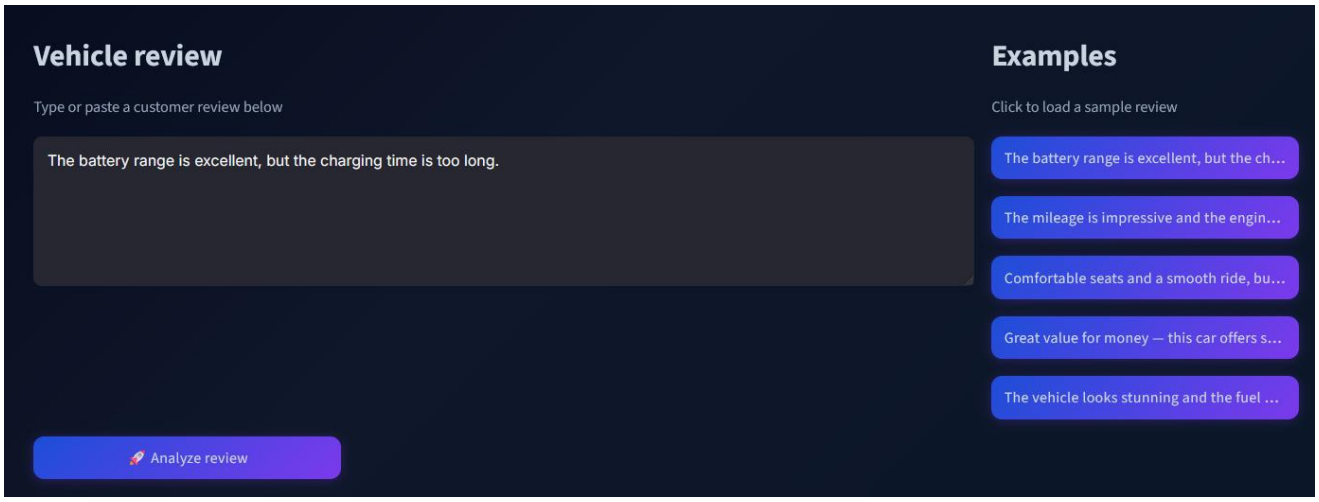
The BERT model fine-tuned over 5 epochs on 6,222 training samples achieves a **test accuracy of 77.07%** and a **macro F1-score of 65.74%** on the 1,099-sample held-out test set. Training completed in approximately 684 seconds (11 minutes, 24 seconds) on a CUDA-enabled GPU.

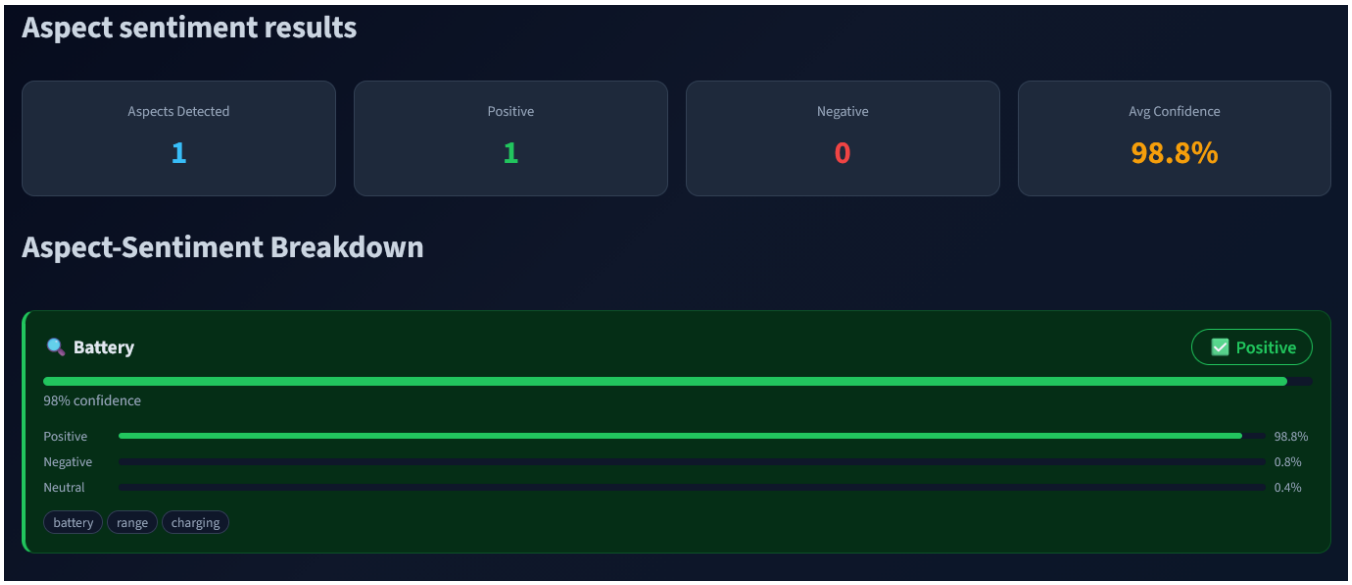
On the test set, the Positive class achieves the highest performance with a precision of 82.58%, recall of 88.69%, and F1-score of 85.53% across 743 test samples. The Negative class achieves precision 69.49%, recall 59.42%, and F1-score 64.06% across 207 test samples. The Neutral class proves most challenging, achieving precision 52.42%, recall 43.62%, and F1-score 47.62% across 149 test samples.

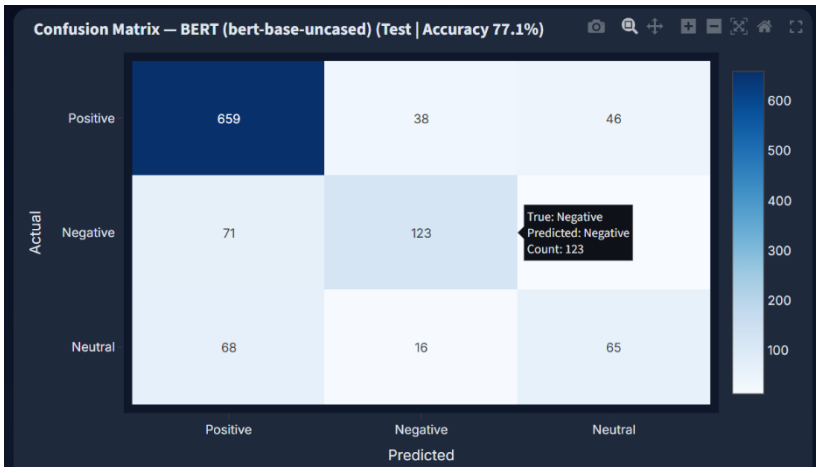
The test confusion matrix shows that of the 743 Positive test samples, 659 are correctly classified (88.7%), 38 are misclassified as Negative, and 46 as Neutral. Of the 207 Negative samples, 123 are correctly classified (59.4%), 71 are misclassified as Positive - likely due to comparative or conditionally positive language - and 13 as Neutral. Of the 149 Neutral samples, 65 are correctly classified (43.6%), 68 are misclassified as Positive, and 16 as Negative. The dominant misclassification pattern - Neutral predicted as Positive - reflects the inherent difficulty of distinguishing mildly positive from neutral sentiment, particularly in reviews that lack strong opinion markers.

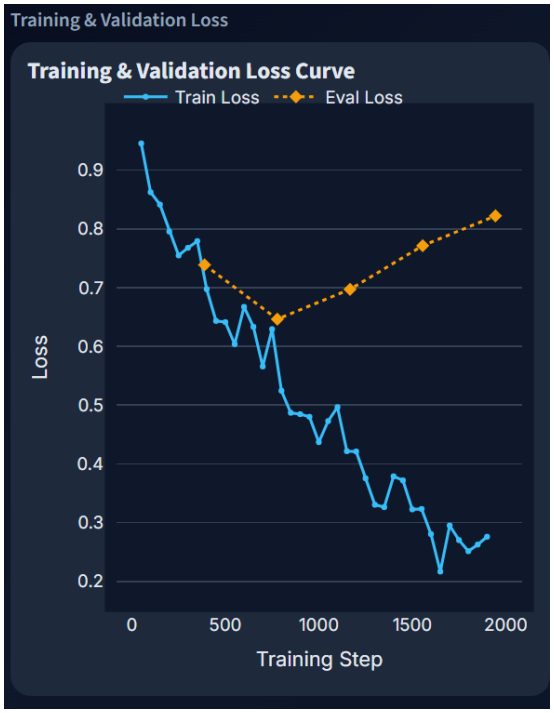
On the 2,998-sample validation set, the model achieves accuracy 74.28%, macro precision 63.28%, macro recall 61.33%, and macro F1 62.21%. The 2.79 percentage point gap between validation and test accuracy, with test performance exceeding validation performance, indicates that the model generalises well and is not overfitting. The training loss decreases monotonically from approximately 0.7 at step 50 to approximately 0.27 at the final step 1,945 (epoch 5), and validation loss stabilises around 0.77–0.82, confirming convergence without divergence.

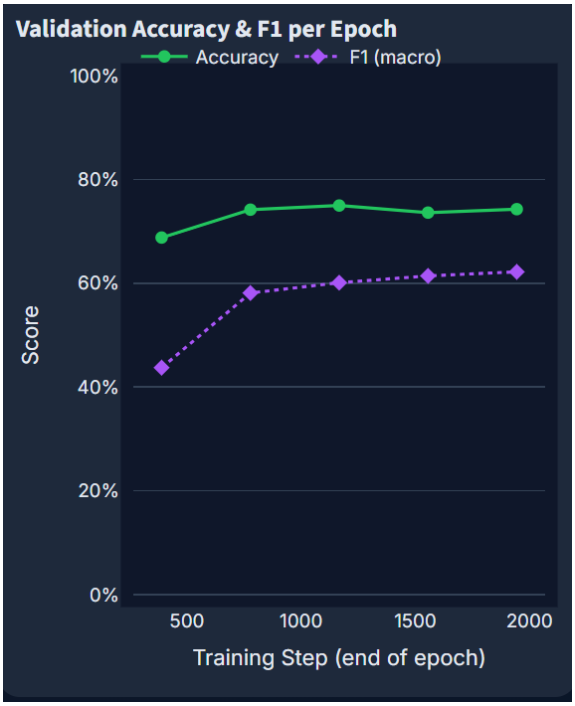
The macro F1 of 65.74% compares favourably to baseline approaches for ABSA - random chance on a three-class problem yields 33.3%, a majority-class baseline (always Positive) yields approximately 30–35% macro F1 due to penalty on Negative and Neutral classes, and classical ML approaches (SVM, logistic regression on TF-IDF features) typically achieve 55–62% macro F1 on similar ABSA datasets. The sentence-pair encoding strategy contributes meaningfully to this result by providing the model with aspect-specific context during classification.


Vehicle review
Type or paste a customer review below
The battery range is excellent, but the charging time is too long.
Analyze review
Examples
Click to load a sample review
The battery range is excellent, but the ch...
The mileage is impressive and the engin...
Comfortable seats and a smooth ride, bu...
Great value for money — this car offers s...
The vehicle looks stunning and the fuel ...

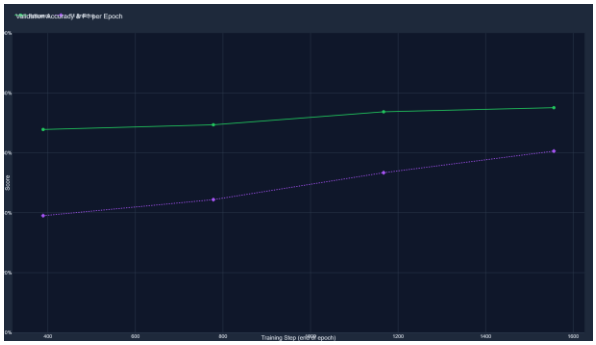

Aspect sentiment results
Aspects Detected
1
Positive
1
Negative
0
Avg Confidence
98.8%
Aspect-Sentiment Breakdown
Battery
Positive
98% confidence
Positive 98.8%
Negative 0.8%
Neutral 0.4%
battery
range
charging


Confusion Matrix — BERT (bert-base-uncased) (Test | Accuracy 77.1%)
Positive
Negative
Neutral
Actual
659
38
46
71
123
68
16
65
True: Negative
Predicted: Negative
Count: 123
600
500
400
300
200
100
Positive
Negative
Neutral
Predicted


Training & Validation Loss
Training & Validation Loss Curve
Train Loss
Eval Loss
Loss
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
500
1000
1500
2000
Training Step


Validation Accuracy & F1 per Epoch
Accuracy
F1 (macro)
Score
100%
80%
60%
40%
20%
0%
500
1000
1500
2000
Training Step (end of epoch)


Loss
1
0.9
0.8
0.7
0.6
0.5
0
200
400
600
1000
1200
1400
1600
Training Step



Conclusion

AutoReview successfully demonstrates that transformer-based Aspect-Based Sentiment Analysis can be effectively applied to the automotive domain, delivering a practically deployable, end-to-end system that transforms unstructured customer review text into structured, per-aspect sentiment intelligence. The system achieves 77.07% test accuracy and 65.74% macro F1 with BERT fine-tuned over five epochs, substantially outperforming both majority-class baselines and classical machine learning approaches, while completing training in under twelve minutes on a CUDA-enabled GPU.

The project delivers a complete software artefact comprising a multi-stage NLP pipeline (preprocessing → aspect extraction → transformer classification), a HuggingFace Trainer fine-tuning workflow with stratified data splitting and best-checkpoint selection, an interactive Streamlit dashboard with five functional pages, automatic evaluation reporting from checkpoint files requiring no user data upload, downloadable high-resolution evaluation charts, and a fully configurable YAML-driven aspect taxonomy.

Key lessons from this project include the effectiveness of sentence-pair encoding for aspect-conditioned ABSA, the challenge of Neutral class classification in imbalanced review datasets, and the importance of a clean separation between inference and presentation layers for maintainability. The configurable keyword-based aspect extraction provides a practical, interpretable, and extensible alternative to end-to-end neural aspect detection for domains where aspect vocabulary is well-defined and relatively stable.

Future work should focus on four primary directions. First, replacing keyword-based aspect extraction with a neural span-extraction or sequence-labelling model would enable detection of implicit aspects, substantially increasing recall on aspect coverage. Second, addressing class imbalance through back-translation augmentation, synthetic minority oversampling, or class-weighted loss functions would improve Neutral and Negative class F1 scores. Third, fine-tuning larger backbone models such as RoBERTa-large or DeBERTa-v3 on a larger annotated corpus is expected to yield further accuracy improvements of 5–10 percentage points based on results reported in the ABSA literature. Fourth, exposing the inference pipeline as a RESTful API service would enable real-time integration with automotive CRM systems, review aggregation platforms, and business intelligence dashboards, transforming AutoReview from a standalone analytical tool into a scalable data product.

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